

# Algorithms for Modern Processor Architectures

Daniel Lemire, professor  
Université du Québec (TÉLUQ)  
Montréal 🇨🇦

blog: <https://lemire.me>

X: [@lemire](#)

GitHub: <https://github.com/lemire/>

All software for this talk: <https://github.com/lemire/talks/tree/master/2025/sea/software>

## Disk at gigabytes per second

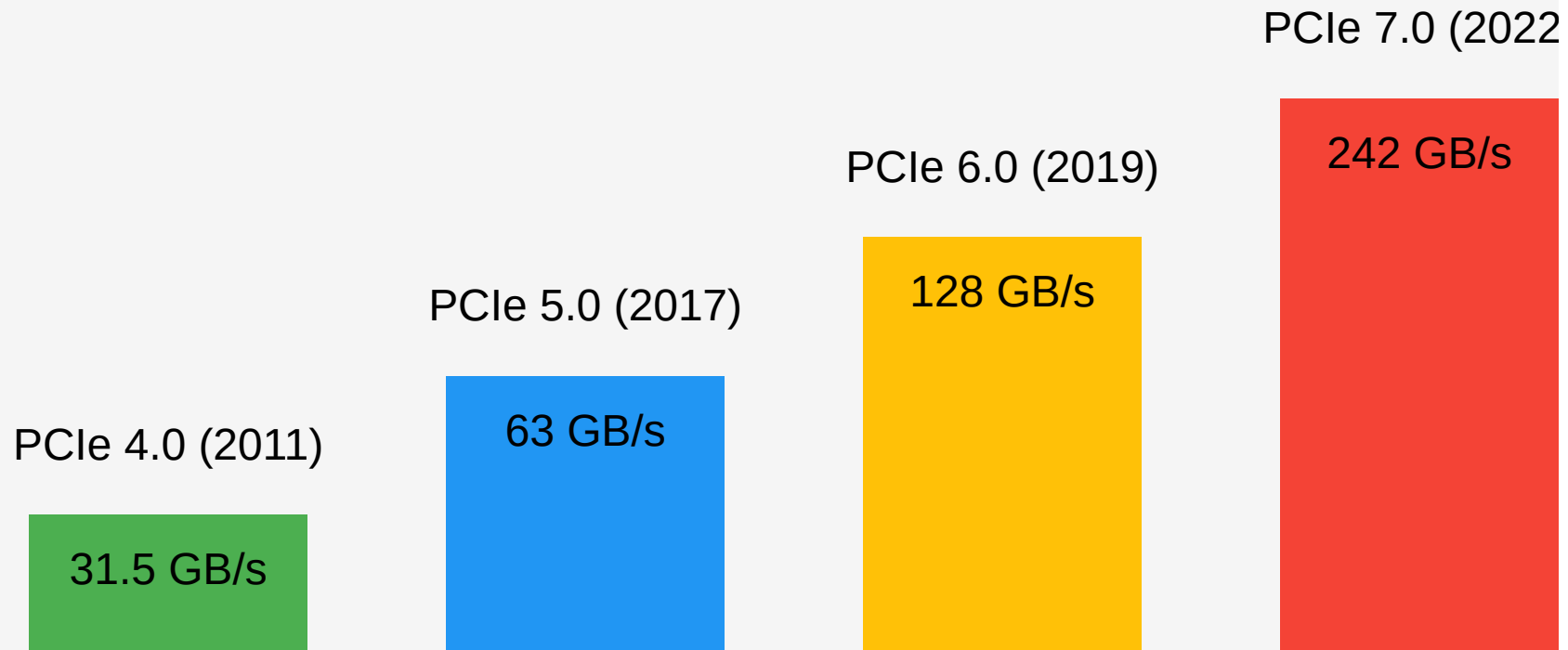
**Micron shows off world's fastest PCIe 6.0 SSD, hitting 27 GB/s speeds — Astera Labs PCIe 6.0 switch enables impressive sequential reads**

News

By [Sunny Grimm](#) published March 8, 2025

The next-gen of networking and storage is hitting the trade shows

## PCI Express Bandwidth Comparison



# High Bandwidth Memory

- Xeon Max processors contain 64 GB of HBM
- Bandwidth 800 GB/s

|           |           |           |           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Wikipedia | Wikipedia | Wikipedia | Wikipedia | Wikipedia | Wikipedia | Wikipedia | Wikipedia |
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# Some numbers

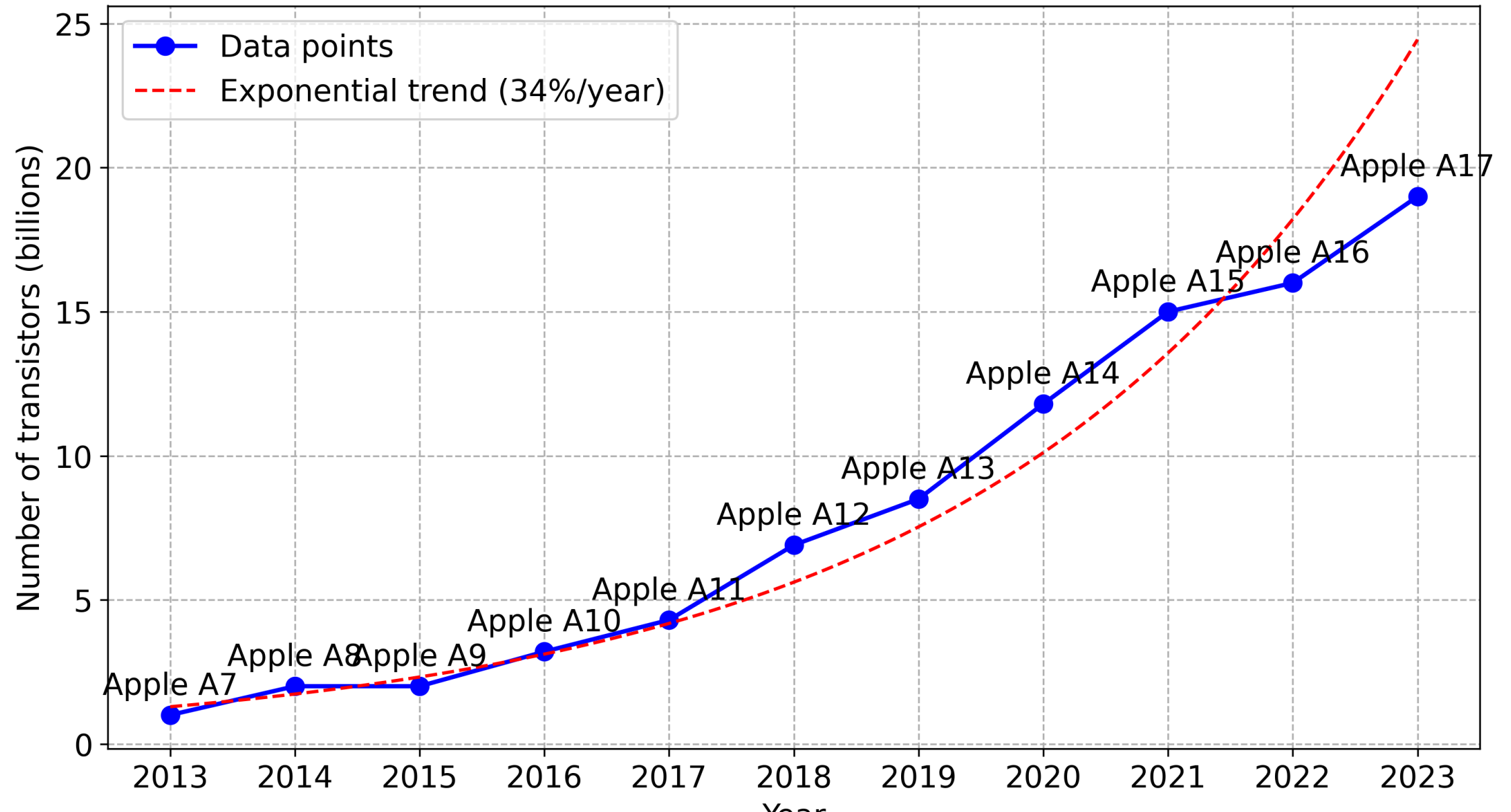
- Time is discrete: clock cycle
- Processors: 4 GHz ( $4 \times 10^9$  cycles per second)
- One cycle is 0.25 nanoseconds
- light: 7.5 centimeters per cycle
- One byte per cycle: 4 GB/s

**Easily CPU bound**

# Frequencies and transistors

| processor     | year | frequency | transistors    |
|---------------|------|-----------|----------------|
| Pentium 4     | 2000 | 3.8 GHz   | 0.040 billions |
| Intel Haswell | 2013 | 4.4 GHz   | 1.4 billions   |
| Apple M1      | 2020 | 3.2 GHz   | 16 billions    |
| Apple M2      | 2022 | 3.49 GHz  | 20 billions    |
| Apple M3      | 2024 | 4.05 GHz  | 25 billions    |
| Apple M4      | 2024 | 4.5 GHz   | 28 billions    |
| AMD Zen 5     | 2024 | 5.7 GHz   | 50 billions    |

Year vs number of transistors





# Where do the transistors go?

- More cores
- More superscalar execution
- Better speculative execution
- More cache, more memory-level parallelism
- Better data-level parallelism (SIMD)

# Where do the transistors go?

- More cores
- More superscalar execution (more instructions per cycle)
- Better speculative execution (→ more instructions per cycle)
- More cache, more memory-level parallelism (→ more instructions per cycle)
- Better data-level parallelism (SIMD) (→ fewer instructions)

# Superscalar execution

| processor       | year | arithmetic logic units | SIMD units     |
|-----------------|------|------------------------|----------------|
| Pentium 4       | 2000 | 2                      | $2 \times 128$ |
| AMD Zen 2       | 2019 | 4                      | $2 \times 256$ |
| Apple M*        | 2019 | 6+                     | $4 \times 128$ |
| Intel Lion Cove | 2024 | 6                      | $4 \times 256$ |
| AMD Zen 5       | 2024 | 6                      | $4 \times 512$ |

Moving to up to 4 load/store per cycle

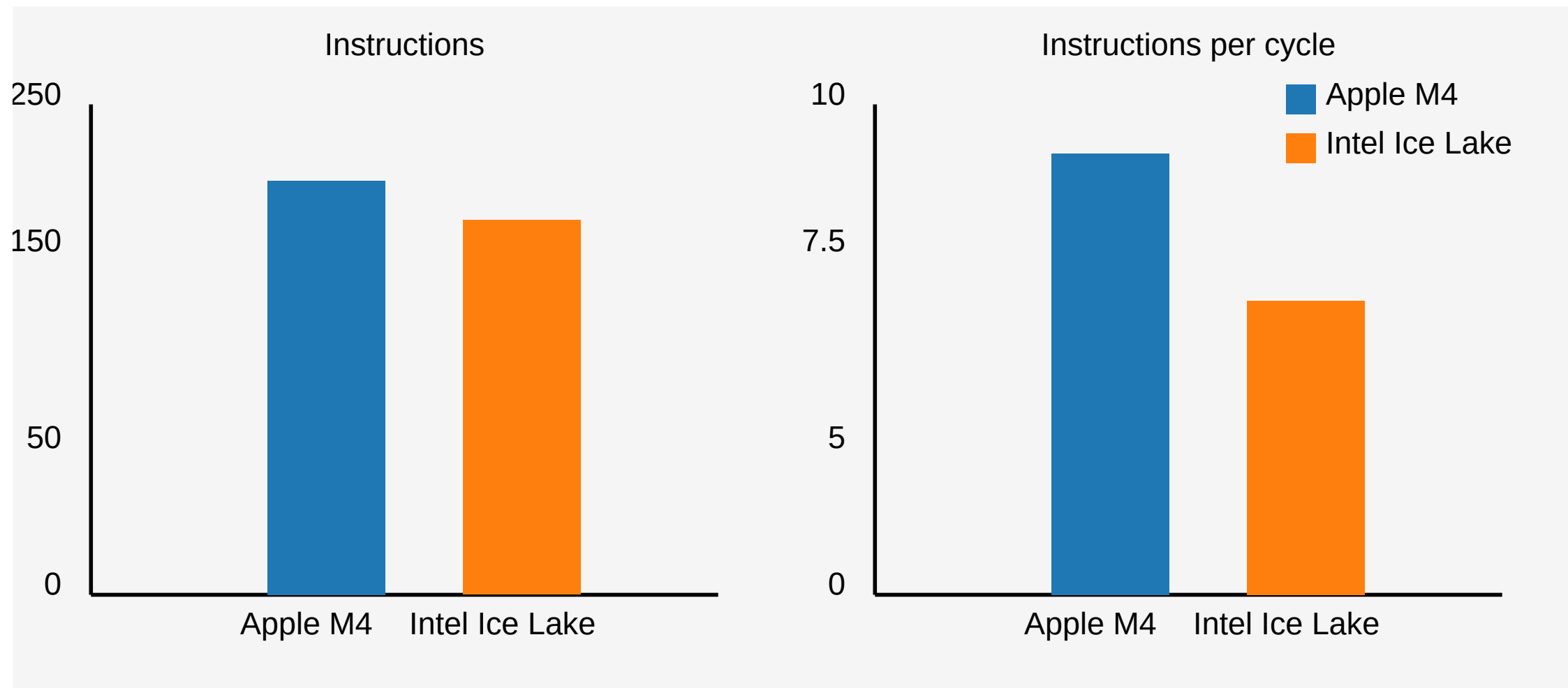
# Parsing a number

- `1.3321321e-12` to `double`

```
double result;  
fast_float::from_chars(  
    input.data(), input.data() + input.size(), result);
```

*Reference:* Number Parsing at a Gigabyte per Second, Software: Practice and Experience 51 (8), 2021

# Parsing a number



# Lemire's Rule 1

Modern processors execute *nearly* as many instructions per cycle as you can supply.

- with caveats: branching, memory, and input/output

# Lemire's Corollary 1

In computational workloads (batches), minimizing instruction count is critical for achieving optimal performance.

## Lemire's Tips

1. Batch your work in larger units to save instructions.
2. Simplify the processing down to as few instructions as possible.



# Going back to number parsing

- Our number parser: major browsers (Safari, Chrome), GCC (12+), C#, Rust
- About  $4\times$  faster than the conventional alternatives.
- How did we do it?

We massively reduced the number of CPU instructions required.

| function   | instructions  |
|------------|---------------|
| strtod     | $> 1000$      |
| our parser | $\approx 200$ |

*Reference:*

Number Parsing at a Gigabyte per Second, Software: Practice and Experience 51 (8), 2021

# SWAR

- Stands for SIMD within a register
- Use normal instructions, portable (in C, C++,...)
- A 64-bit register can be viewed as 8 bytes
- Requires some cleverness

## Check whether we have a digit

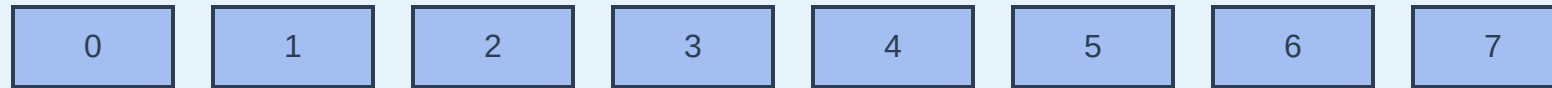
In ASCII/UTF-8, the digits 0, 1, ..., 9 have values 0x30, 0x31, ..., 0x39.

To recognize a digit:

- The high nibble should be 3.
- The high nibble should remain 3 if we add 6 (0x39 + 0x6 is 0x3f)

## ASCII processing flow

ASCII numbers



ASCII codes (hex)



Zero lower 4 bits (AND 0xF0)



Add 6, shift right 4



OR result (hex)



# Batching (unrolling)

6 to 7 instructions per multiplication

```
for (size_t i = 0; i < length; i++)  
    sum += x[i] * y[i];
```

3 to 5 instructions per multiplication

```
for (; i < length - 3; i += 4)  
    sum += x[i] * y[i]  
           + x[i + 1] * y[i + 1]  
           + x[i + 2] * y[i + 2]  
           + x[i + 3] * y[i + 3];
```

## Apple M4

Instructions



Regular: 6



Unrolled: 3.5

Cycles



Regular: 1.2



Unrolled: 1.0

## Intel Ice Lake

Instructions



Regular: 7

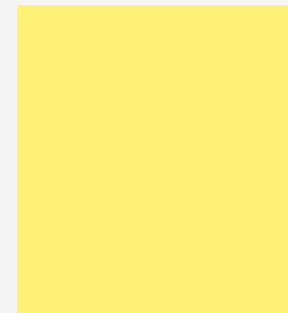


Unrolled: 5

Cycles



Regular: 1.6



Unrolled: 1.1

# Knuth's random shuffle

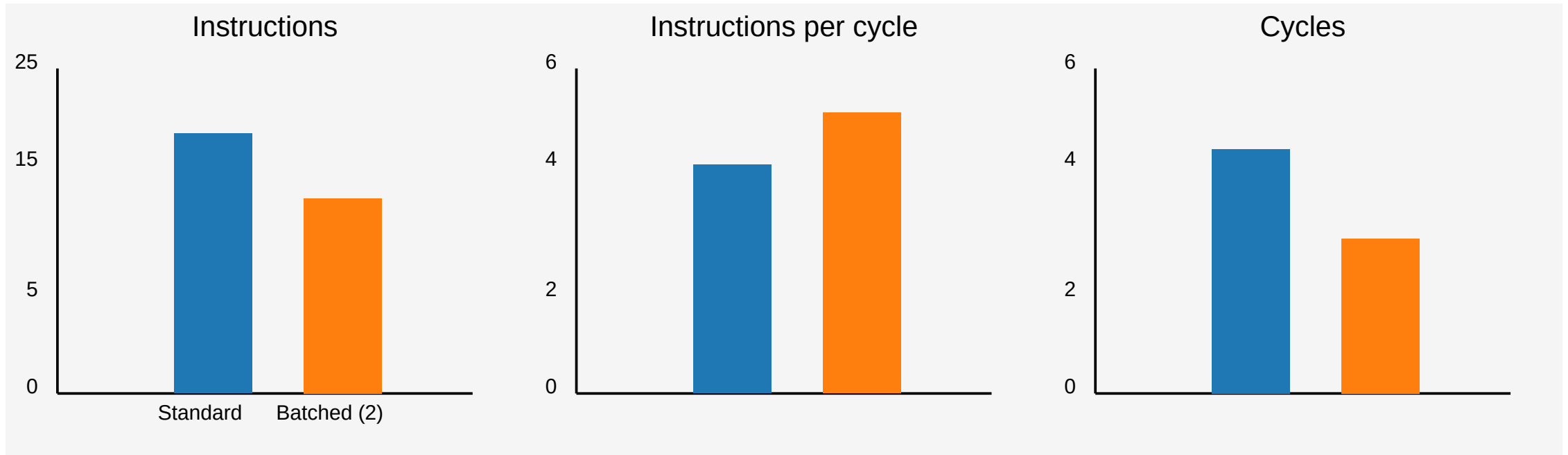
```
PROCEDURE shuffle(array)
  FOR j FROM |array| - 1 DOWN TO 1
    k ← random_integer(0, j)
    SWAP array[j] WITH array[k]
  END FOR
END PROCEDURE
```



# Batched random shuffle

- Draw one random number
- Compute two indices (with high probability)
- Reduces the instruction count
- Reduces the number of branches

## Results (Apple M4): Use a large array (8 MB).



*Reference:* Batched Ranged Random Integer Generation, Software: Practice and Experience 55 (1), 2025

# Branching

Hard-to-predict branches can derail performance

# Unicode (UTF-16)

- Code points from U+0000 to U+FFFF, a single 16-bit value.
- Beyond: a surrogate pair [U+D800 to U+DBFF] followed by U+DC00 to U+DFFF

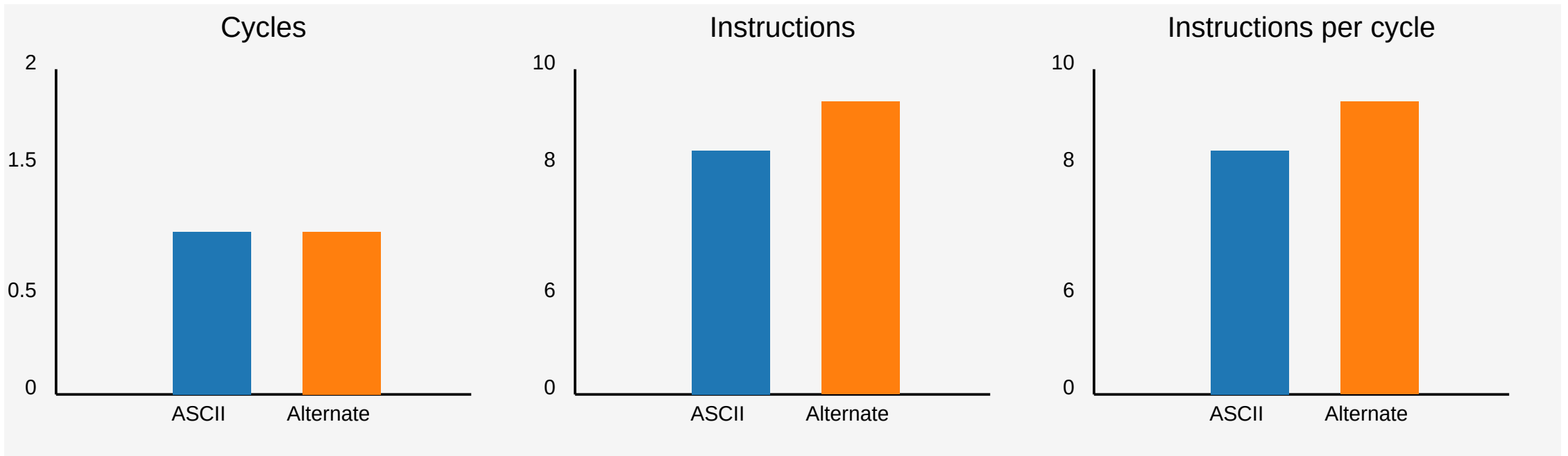
# Validate

- Check for a lone code unit ( $x \leq 0xD7FF \vee x \geq 0xDBFF$ ), if so ok
- Check for the first part of the surrogate ( $0xD800 \leq x \leq 0xDBFF$ ) and if so check that we have the second part of a surrogate

# Validate

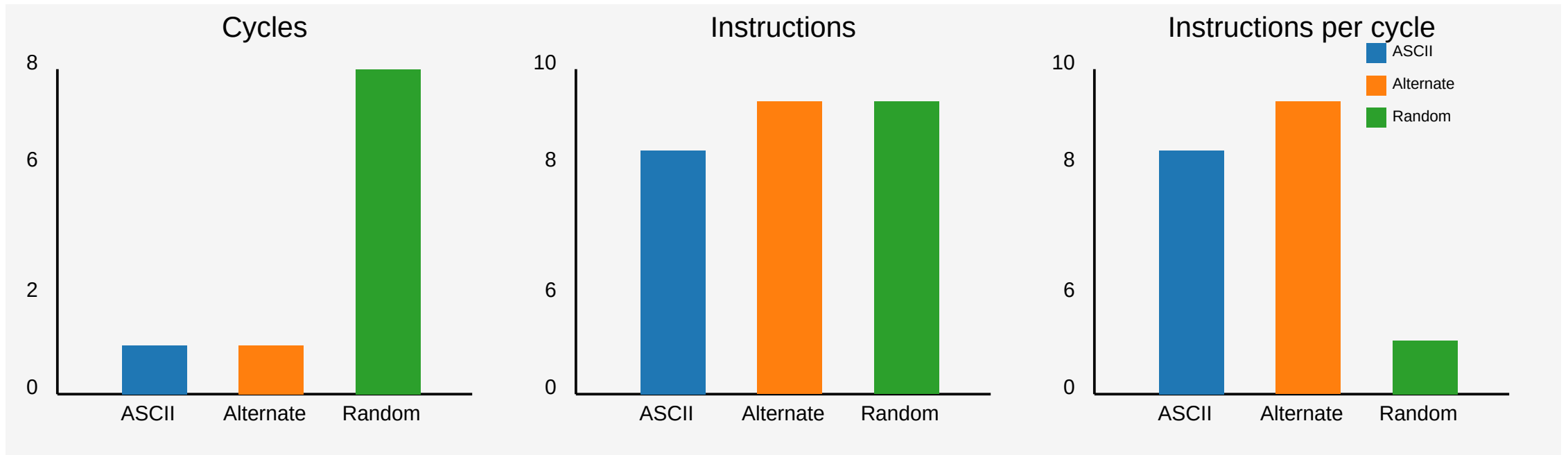
```
PROCEDURE validate_utf16(code_units)
  i ← 0
  WHILE i < |code_units|
    unit ← code_units[i]
    IF unit ≤ 0xD7FF OR unit ≥ 0xE000 THEN
      INCREMENT i
      CONTINUE
    IF unit ≥ 0xD800 AND unit ≤ 0xDBFF THEN
      IF i + 1 ≥ |code_units| THEN
        RETURN false
      next_unit ← code_units[i + 1]
      IF next_unit < 0xDC00 OR next_unit > 0xDFFF THEN
        RETURN false
      i ← i + 2 // Valid surrogate pair
      CONTINUE
    RETURN false
  RETURN true
```

# Performance results (Apple M4)



1 character per cycle might be just 4 GB/s (slower than disk)

# Performance results (Apple M4)



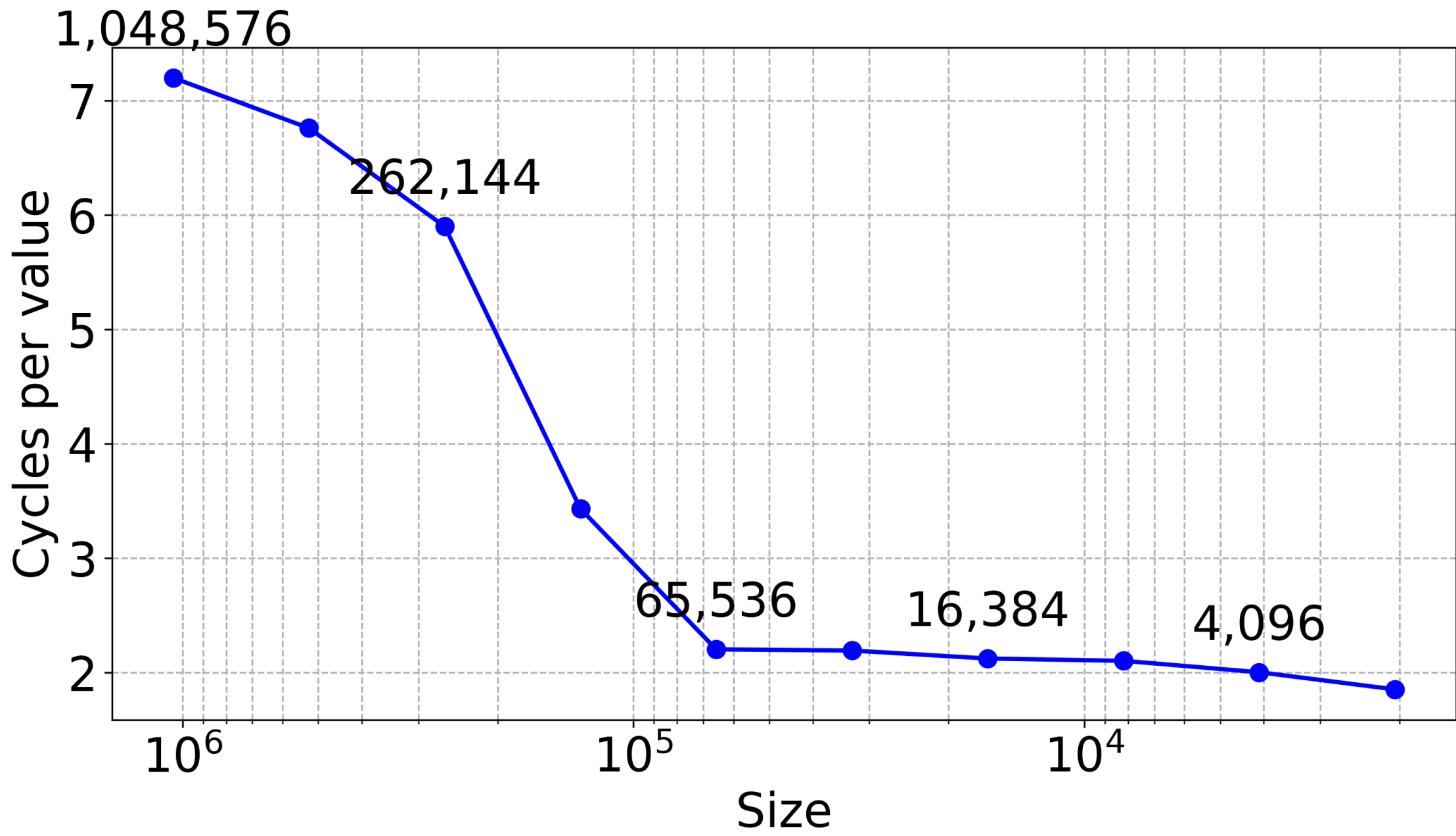
We are now barely at 1 GB/s!



# Speculative execution

- Processors *predict* branches
- They execute code *speculatively* (can be wrong!)

**How much can your processor learn?**



# Finite state machine to the rescue

- Can identify characters by the most significant 8 bits.
- Trivial finite state machine: default, has just encountered a high surrogate, or error.

```

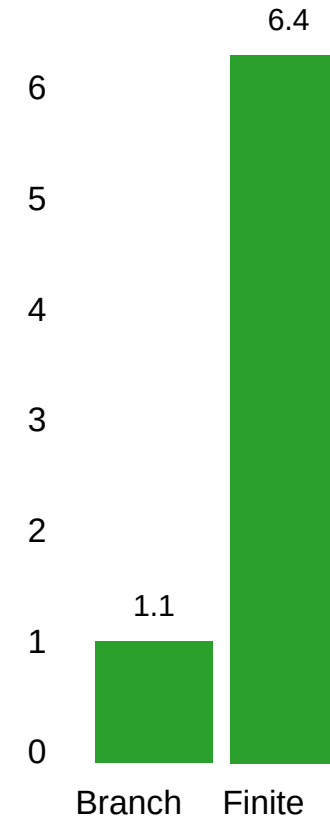
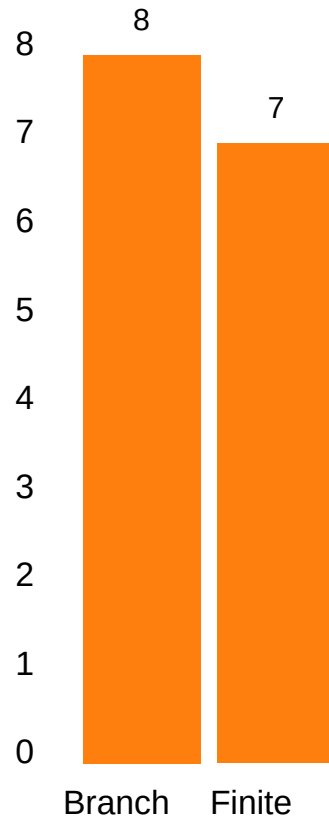
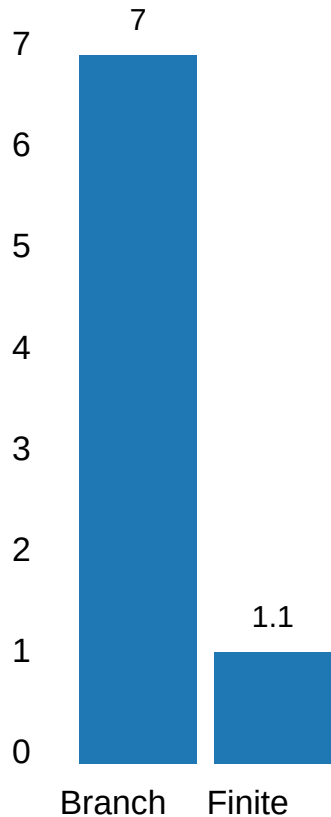
static uint8_t transition_table[3][256] = {
    {...},
    {...},
    {...}
};

bool is_valid_utf16_ff(std::span<uint16_t> code_units) {
    uint8_t state = 0; // Start in Initial state
    for (auto code_unit : code_units) {
        uint8_t high_byte = code_unit >> 8;
        state = transition_table[state][high_byte];
    }
    return state == 0; // Valid only if we end in Initial state
}

```

# Performance results (Apple M4)

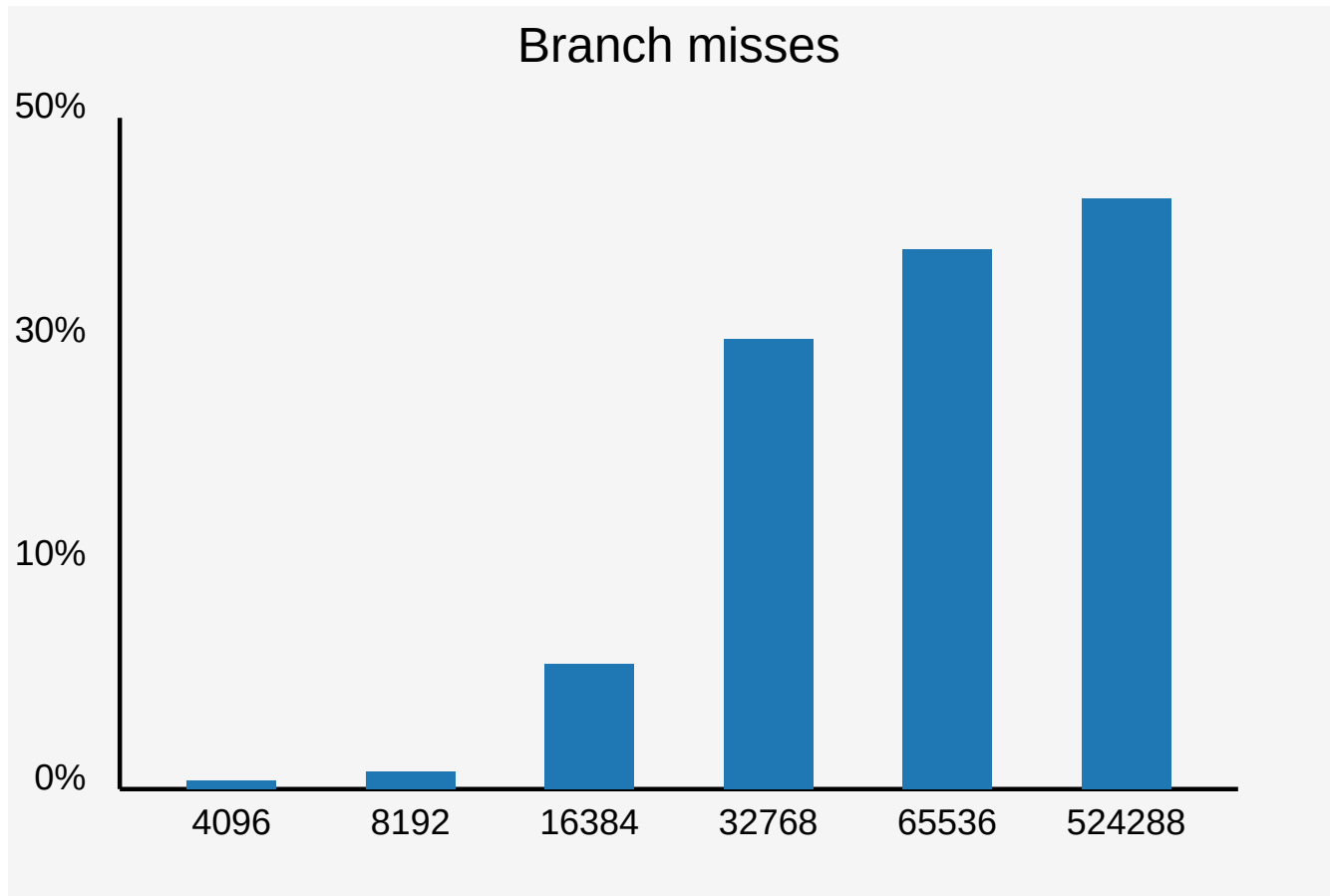
The finite-state approach can be  $7\times$  faster!



# Rules of thumb

1. Processors can 'learn' thousands of branches: benchmark over massive inputs.
2. Pick a solution without branches when it provides the same performance.

# Apple M4 can learn 10,000 random (0/1) branches.





# Pipelining

How does the processor manage to validate one UTF-16 character per cycle when it takes **many cycles** just to *load* the character?

| cycle | action        | action        | pizza en route |
|-------|---------------|---------------|----------------|
| 1     | order pizza A |               |                |
| 2     | order pizza B |               | A 🚚            |
| 3     | order pizza C |               | A 🚚, B 🚚       |
| 4     | order pizza D | eat pizza A 🍕 | B 🚚, C 🚚       |
| 5     | order pizza E | eat pizza B 🍕 | C 🚚, D 🚚       |
| 6     | order pizza F | eat pizza C 🍕 | D 🚚, E 🚚       |

# Little's Law

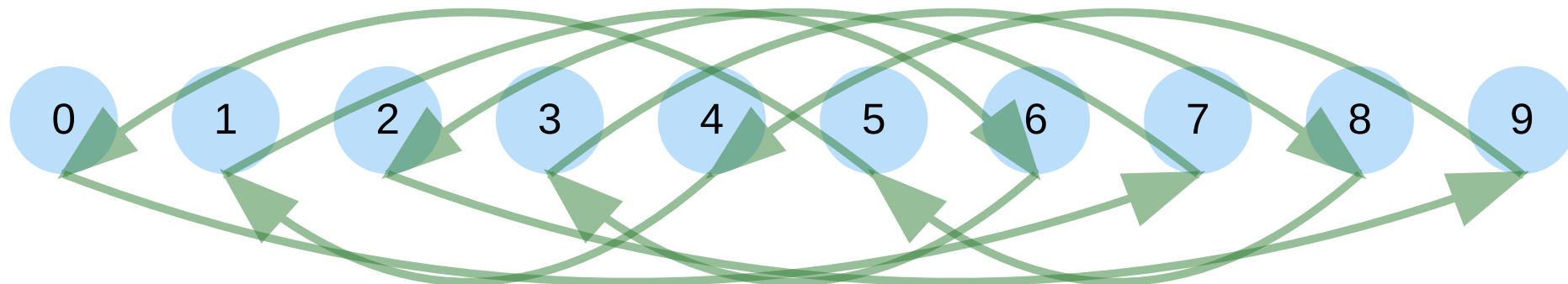
- Latency harms throughput
- Parallelism hides latency

$$\text{throughput} = \frac{\text{parallelism}}{\text{latency}}$$

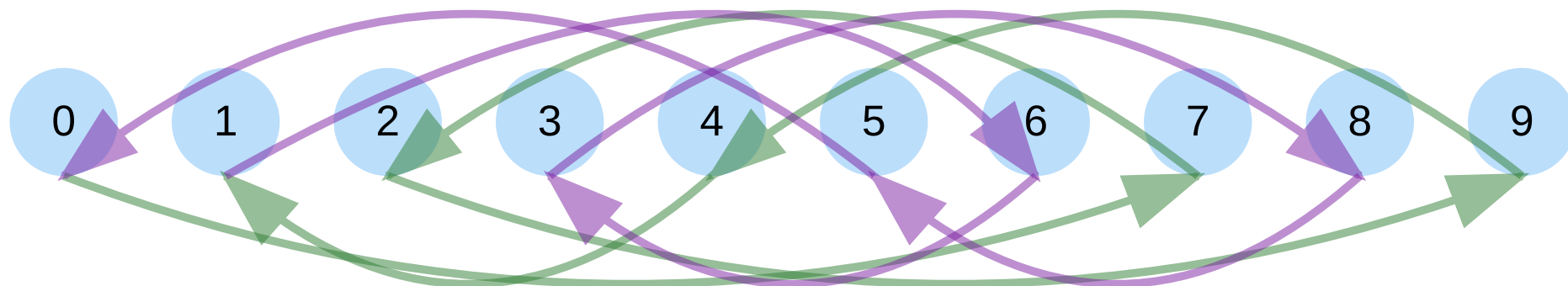
# Memory-level parallelism

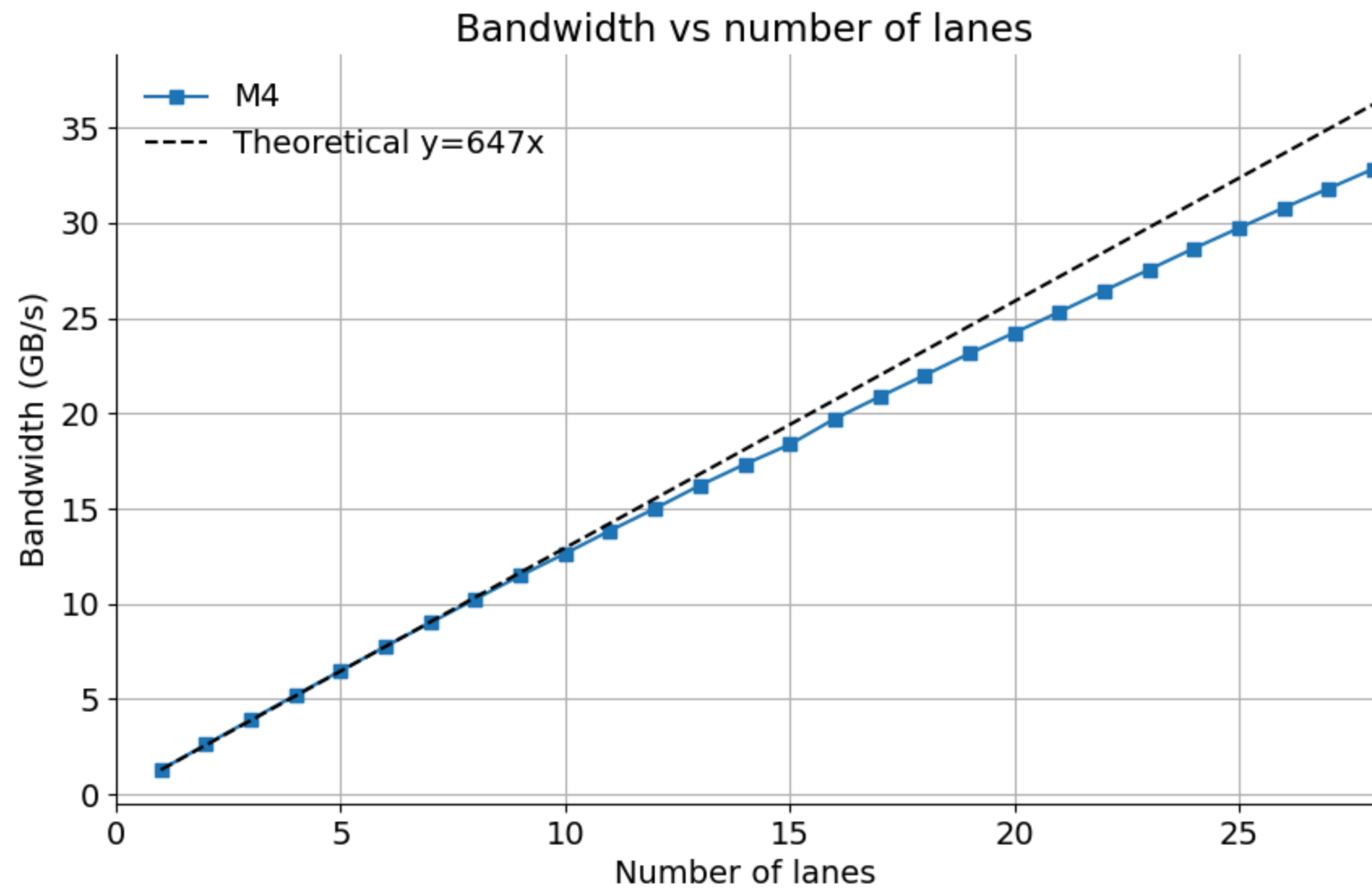
- Create large array of indices forming a cycle
- Start with [0,1,2,3,4]
- Shuffle so that no index can remain in place.
- Start with [4,0,3,2,1]
- Sandra Sattolo's algorithm
- This forms a random path

# 1 lane

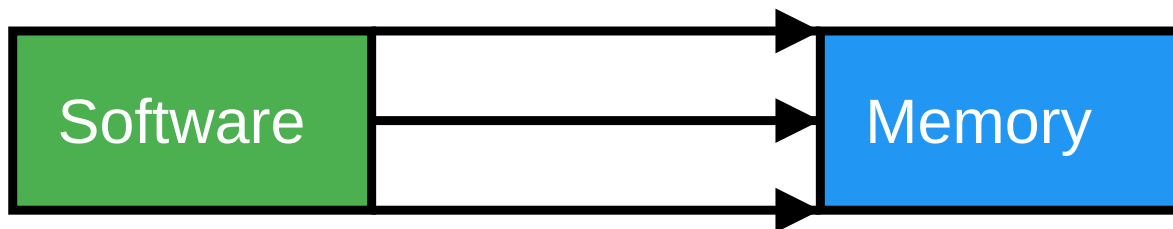
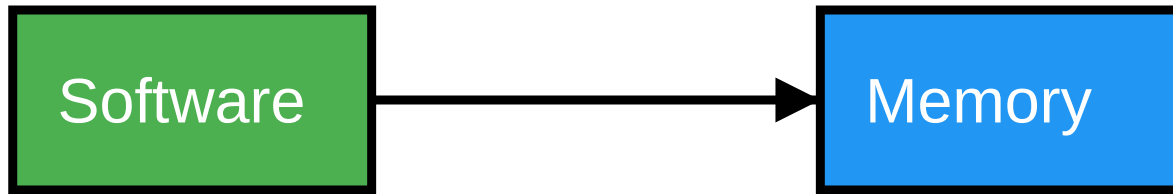


# 2 lanes



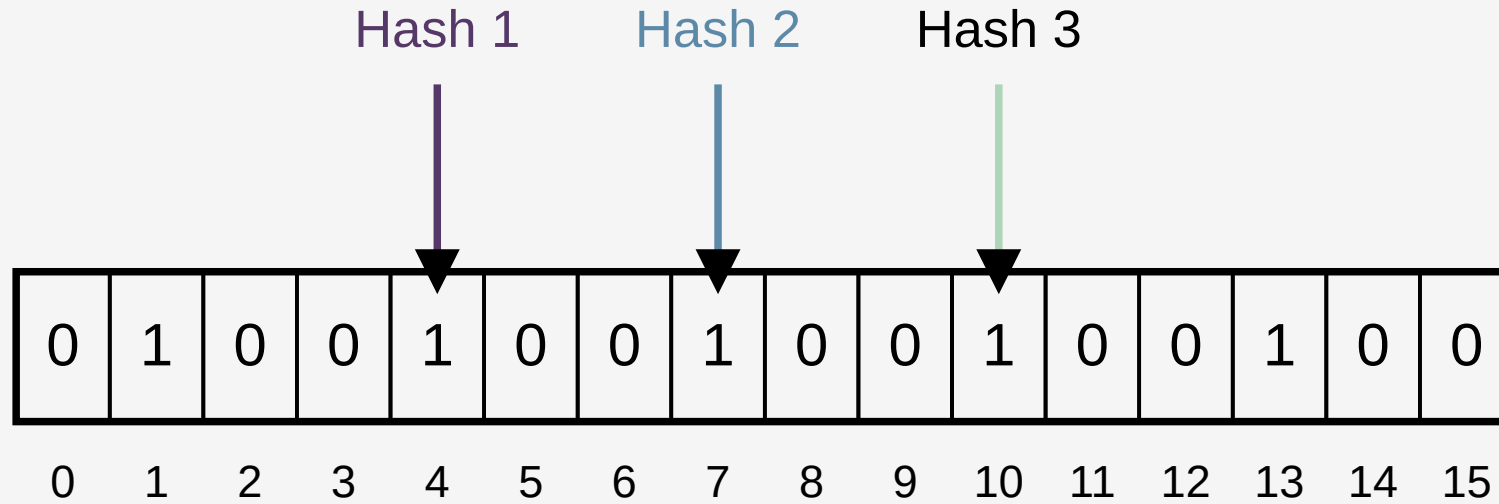


# Consequence



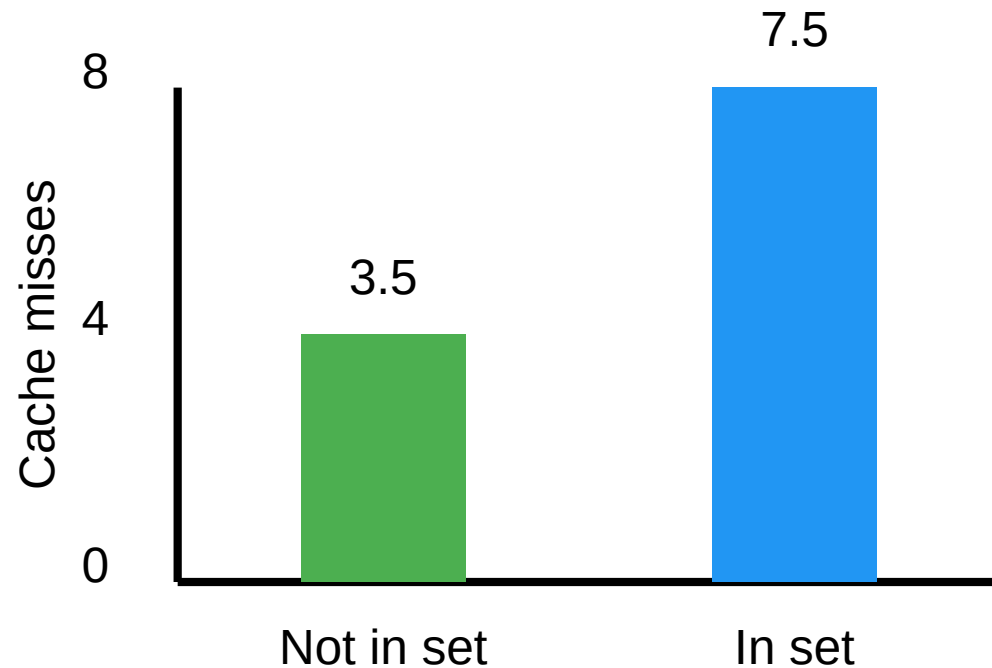


# Bloom filter



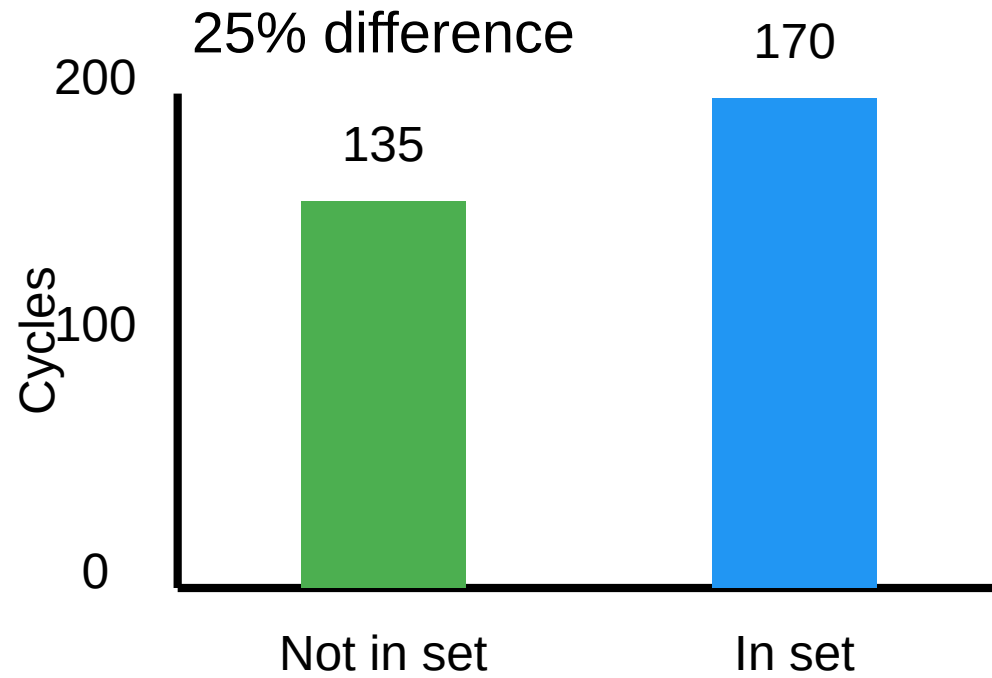
# Bloom filter

8 hash functions, (Intel Ice Lake processor, out-of-cache filter)



Less than half the cache misses

# Bloom filter



# Data-level parallelism

# SIMD

- Stands for Single instruction, multiple data
- Allows us to process 16 bytes or more with one instruction
- Supported on all modern CPUs (phone, laptop)

# ASCII to lower case

```
For each character c
  If c - 'A' < 'Z' - 'A' then
    c = c + 'a' - 'A'
  EndIf
EndFor
```

# ASCII to lower case: 64 characters in 3 instructions

- Compute  $c - A$

```
__m512i ca = _mm512_sub_epi8(c, _mm512_set1_epi8('A'));
```

- Turn  $c - 'A' \leq Z - A$  into a mask

```
__mmask64 is_upper = _mm512_cmple_epu8_mask(ca, _mm512_set1_epi8('Z' - 'A'));
```

- Add  $a - A$  to  $c$  according to mask

```
__m512i to_lower = _mm512_mask_add_epi8(c, is_upper, c, to_lower)
```

# Deltas (C)

successive difference:

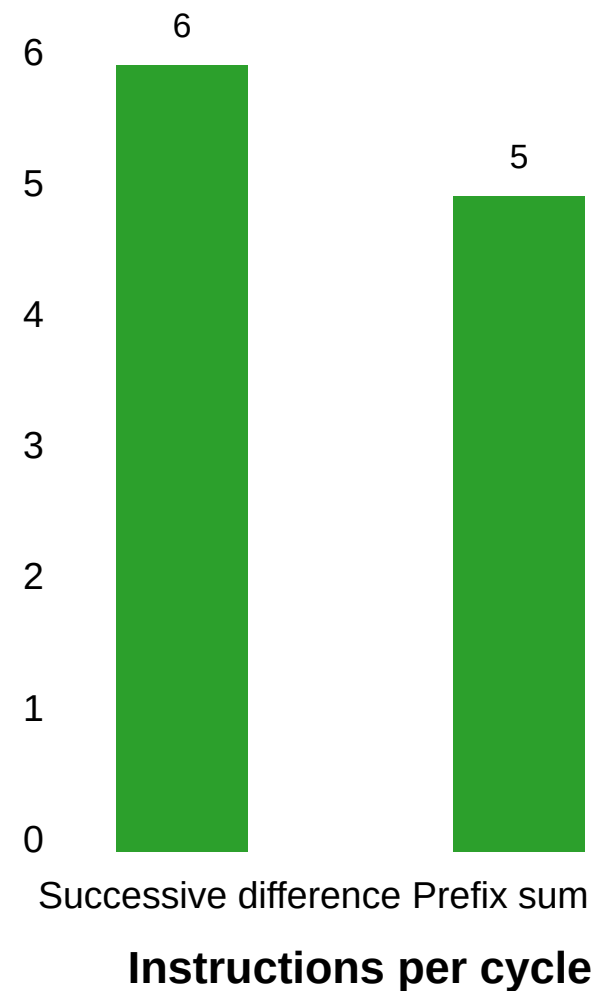
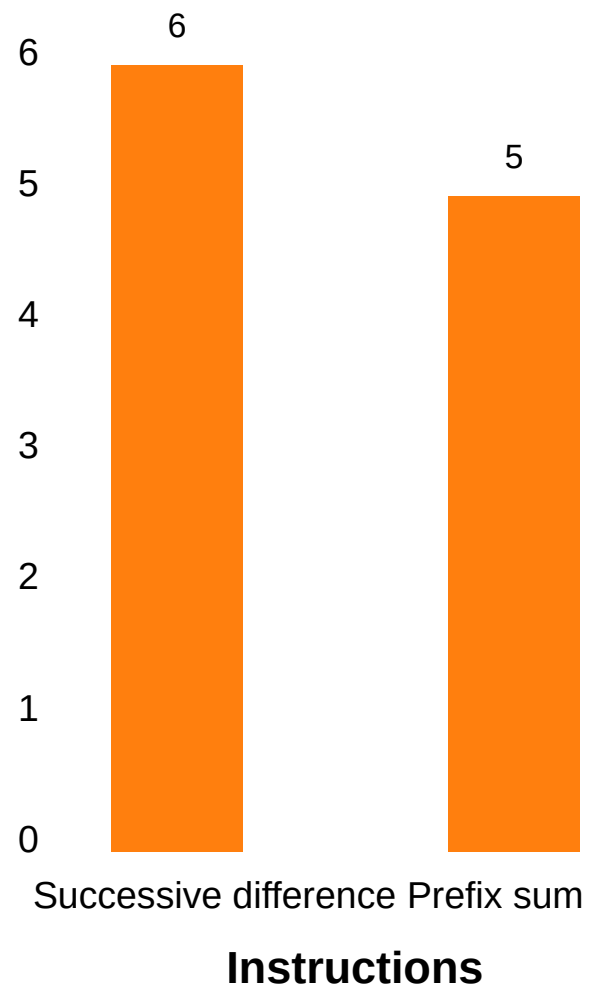
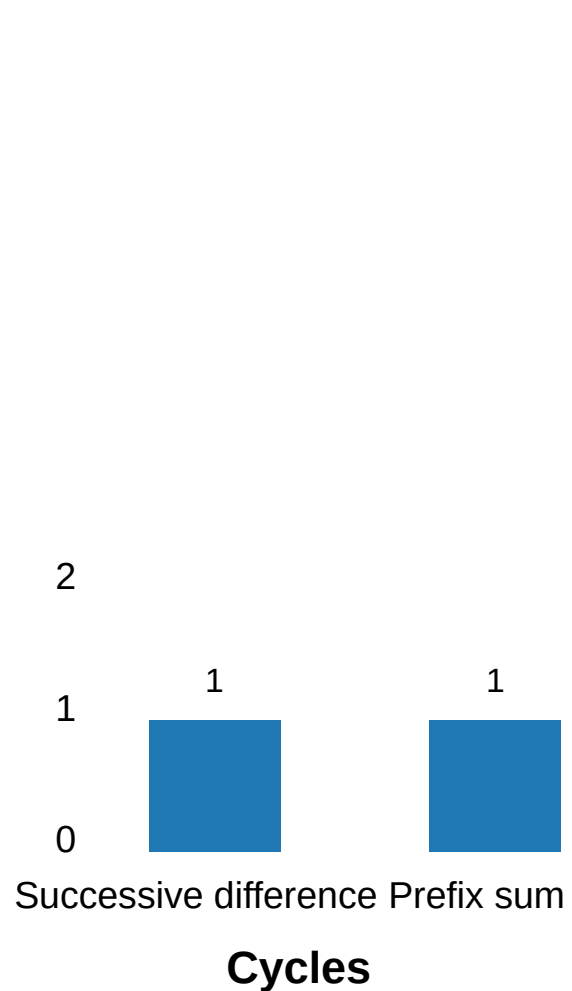
```
for (size_t i = 1; i < n; ++i) {  
    dst[i] = src[i] - src[i - 1];  
}
```

prefix sum:

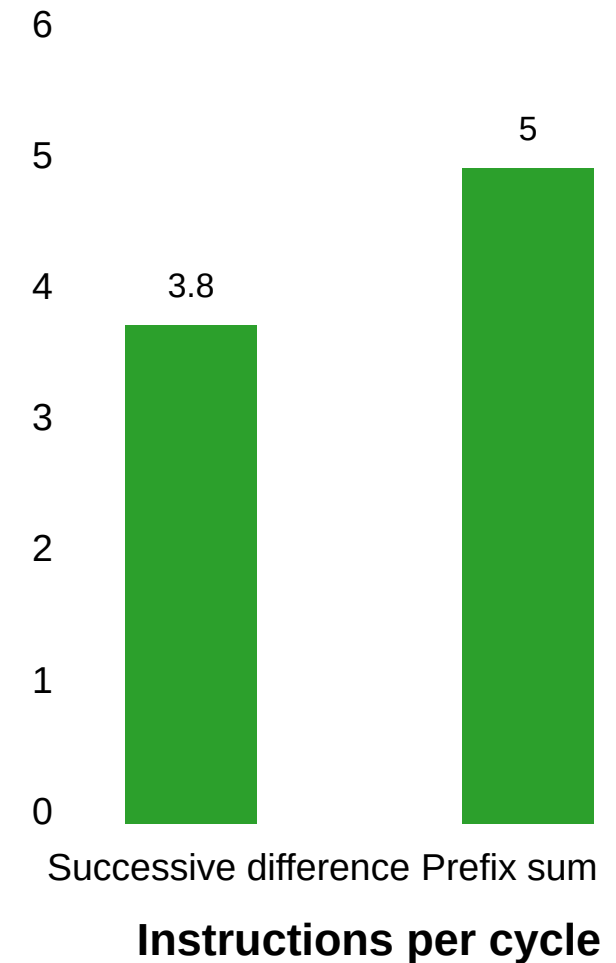
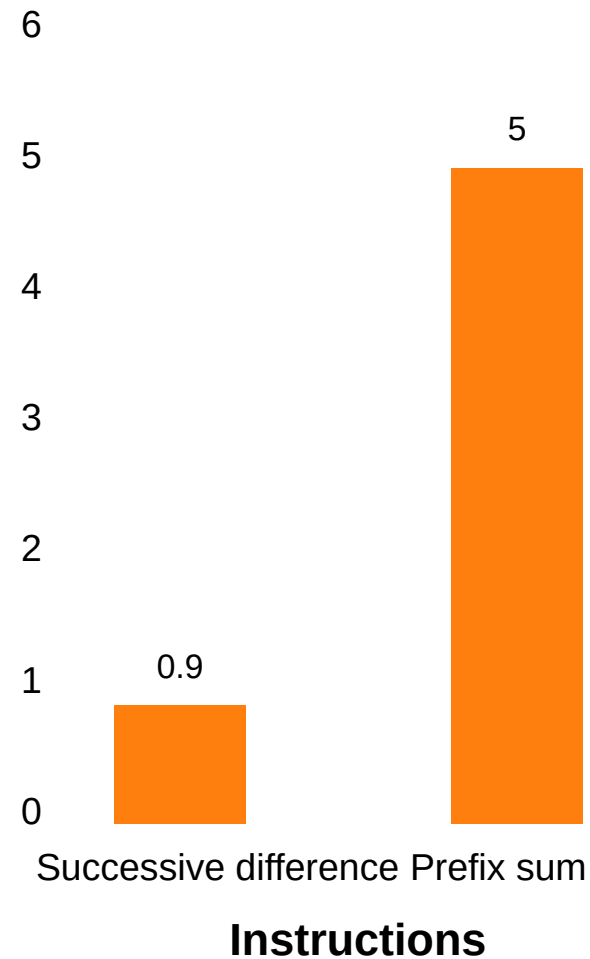
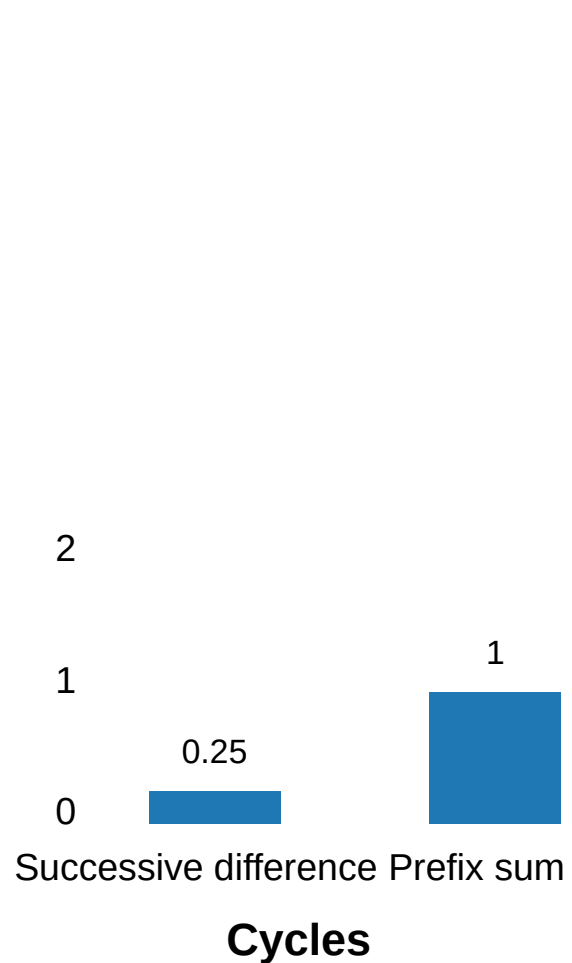
```
for (size_t i = 1; i < n; ++i) {  
    dst[i] = dst[i - 1] + src[i];  
}
```



# Apple M4



# Now allow SIMD! (Autovectorization)

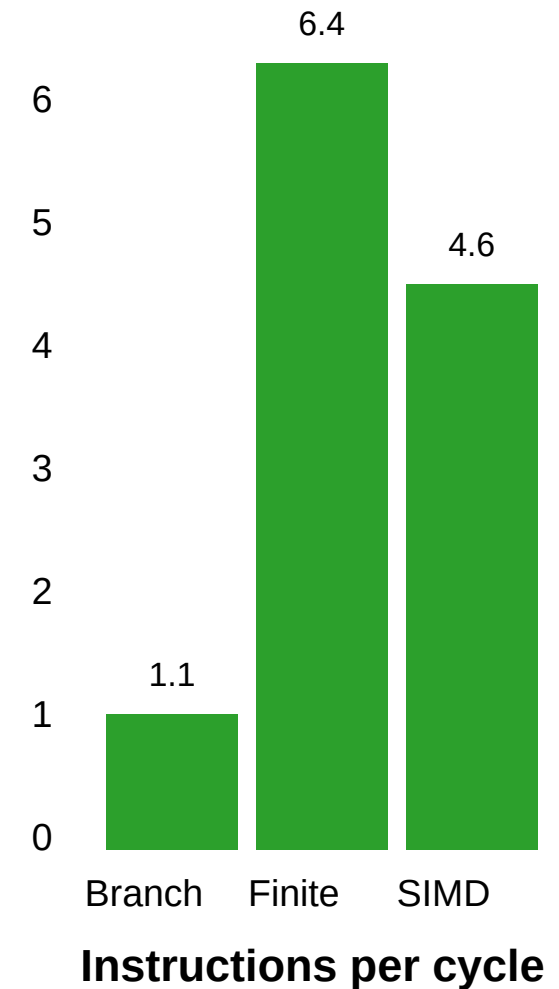
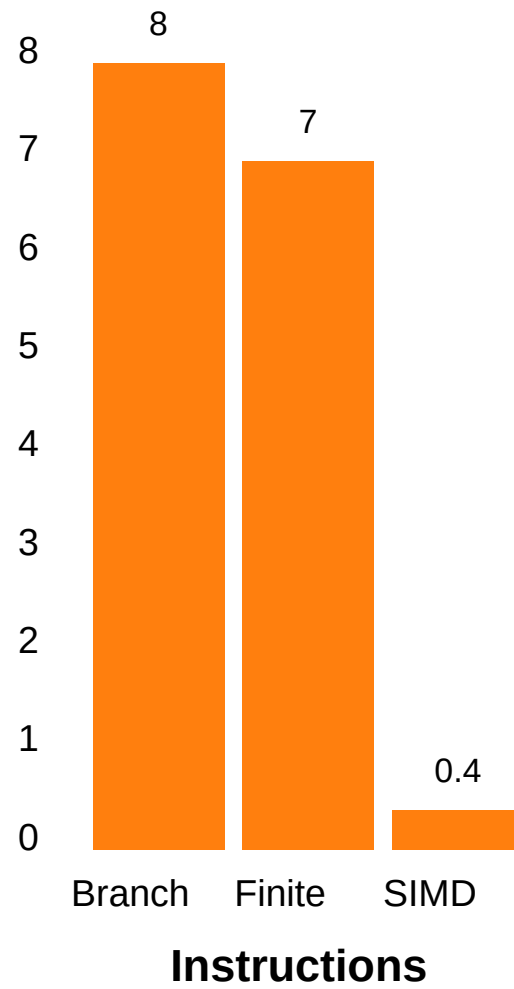
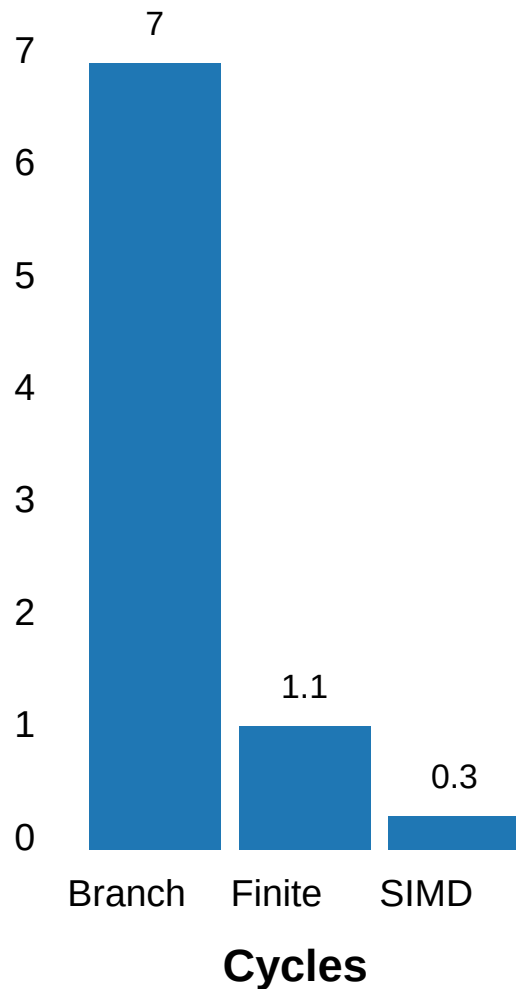


**Need to learn SIMD design magic !**

# UTF-16

- Write SIMD correction function (not just validation)
- Actually deployed in v8 (Google Chrome, Microsoft Edge)

# UTF-16, random (adversarial), Apple M4



- Most x64 processors have AVX2 (32-byte register)
- AMD Zen 5 has powerful AVX-512 (64-byte register)
- ARM has NEON + SVE/SVE2
- RISC-V has its vector instructions
- Loonson processes have AVX2-like instructions

## Interested? Check these projects

- simdjson: The fastest JSON parser in the world <https://simdjson.org>
- simdutf: Unicode routines (UTF8, UTF16, UTF32) and Base64  
<https://github.com/simdutf/simdutf>

# Measurements

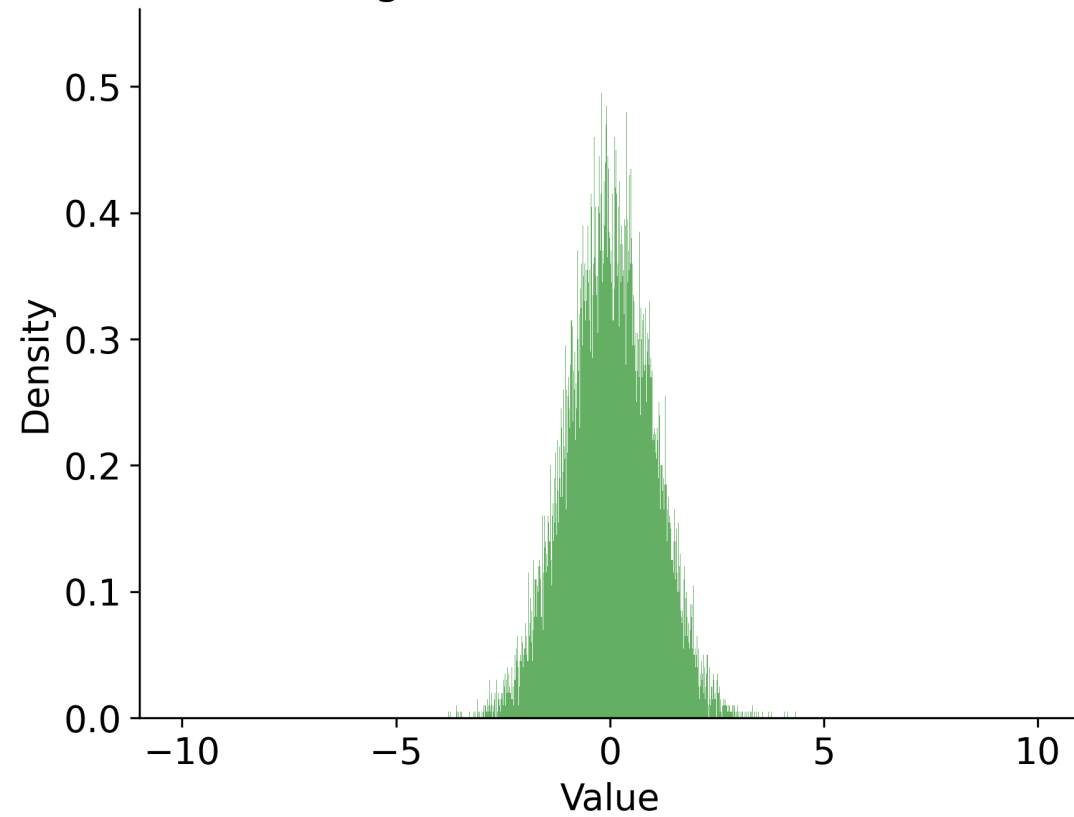
- We often assume that measurements (timings) are normally distributed.
- It is often an incorrect assumption.



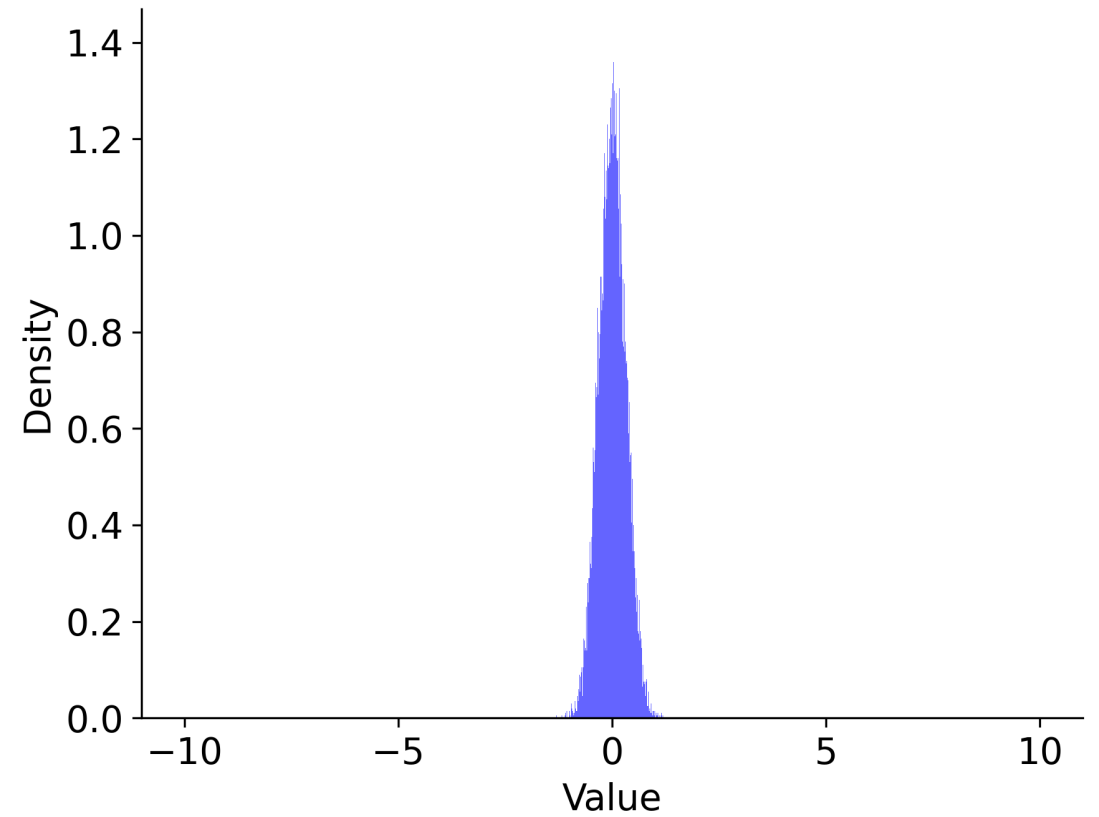
# Measurements

- If your measurements are normally distributed, the 'error' falls off as  $1/\sqrt{N}$

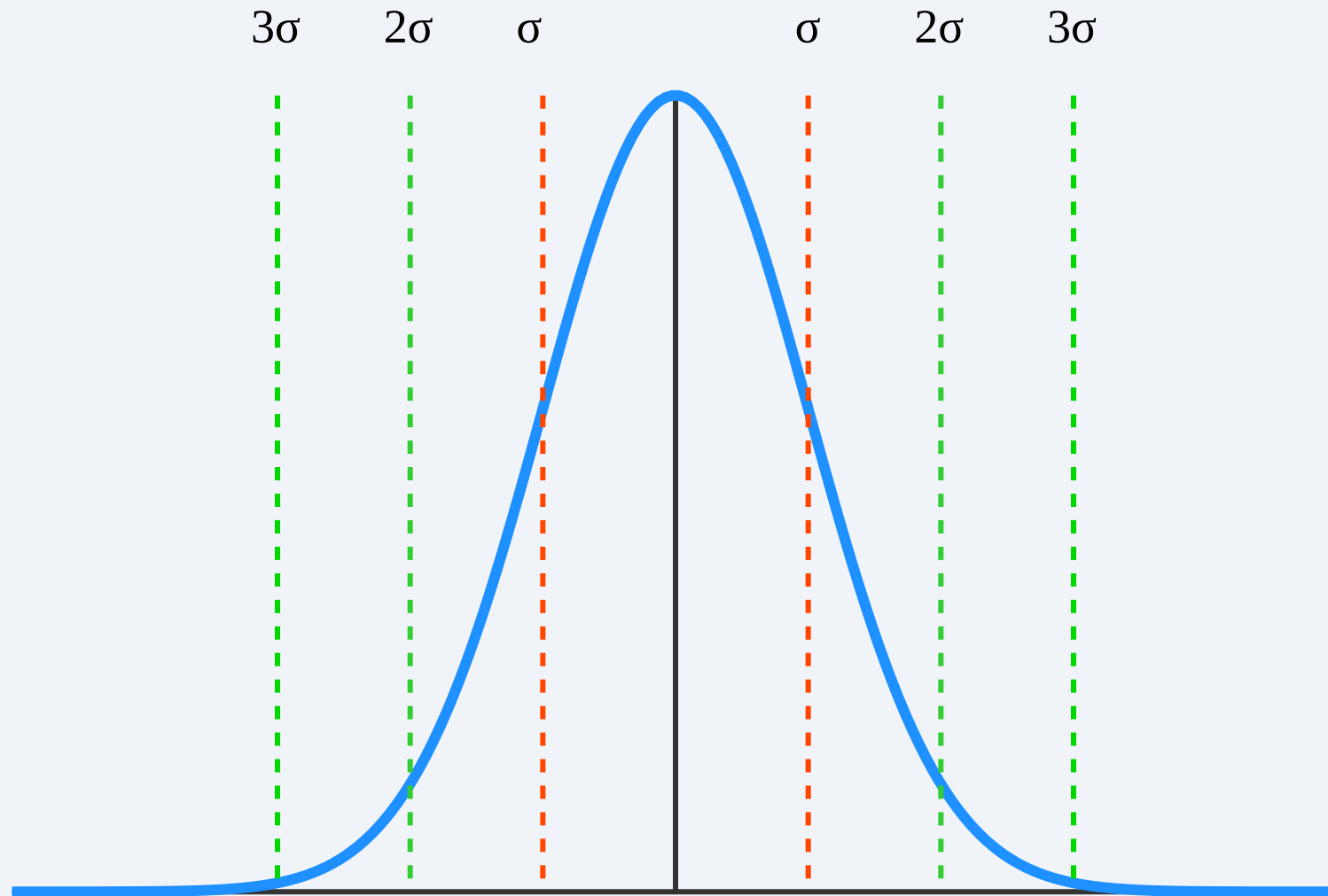
Original normal distribution



Sum of 10 i.i.d. normals / 10



# Sigma events



- 1-sigma is 32%

- 1-sigma is 32%
- 2-sigma is 5%

- 1-sigma is 32%
- 2-sigma is 5%
- 3-sigma is 0.3% (once every 300 trials)

- 1-sigma is 32%
- 2-sigma is 5%
- 3-sigma is 0.3% (once every 300 trials)
- 4-sigma is 0.00669% (once every 15000 trials)

- 1-sigma is 32%
- 2-sigma is 5%
- 3-sigma is 0.3% (once every 300 trials)
- 4-sigma is 0.00669% (once every 15000 trials)
- 5-sigma is 5.9e-05% (once every 1,700,000 trials)
- 6-sigma is 2e-07% (once every 500,000,000)
- $e^{-n^2/2} / (n * \sqrt{\pi/2}) \times 100$  for  $n > 3$



**What if we dealt with log-normal distributions?**

# Real-world measurements

- You cannot assume normality
- Measurements are **not independent**.
- Reality: the absolute minimum is often a *reliable metric*
- Margin: difference between mean and minimum

# Conclusion

- Processors are getting much better! Wider!
- 'hot spot' engineering can fail, better to reduce overall instruction count.
- Branchy code can do well in synthetic benchmarks, but be careful.